# Fundamentals in AI & ML

Course Code: CSA2001

## Practical Task

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Branch: AI & ML

Slot: C14+E11+E12

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# Task 5a (Single Layer Perceptron for Binary Classification)

## Aim

To study and implement Single Layer Perceptron for Binary Classification.

## Apparatus

- Computer or Laptop
- Web Browser
- Internet Connection
- Virtual Labs Platform

## Theory

A Single Layer Perceptron (SLP) is one of the simplest types of artificial neural networks used for binary classification problems. It consists of an input layer and an output layer, where each input is associated with a weight.

The perceptron computes a weighted sum of inputs and passes it through an activation function to produce the output. It works effectively only for linearly separable data.

The main components of a perceptron are:
• Inputs and Weights
• Bias
• Summation function
• Activation function

The output is calculated using:
**Yin = Σ (wi * xi) + bias**

Activation function:

**$f(y) = 1$ if $Y_{in} \geq \theta$**

**$f(y) = -1$ if $Y_{in} < \theta$**

The perceptron learning rule updates weights as:

**$\Delta w = \eta * (\text{target} - \text{output}) * \text{input}$**

# Procedure

Step 1: Open the Virtual Labs website

Step 2: Select “Single Layer Perceptron” experiment

Step 3: Enter input values and parameters

Step 4: Click on Run/Start button

Step 5: Observe the output and classification

Step 6: Repeat for different inputs

# Algorithm / Working:

Step 1: Initialize weights and bias with small values

Step 2: Provide input values and desired output

Step 3: Calculate weighted sum (Yin)

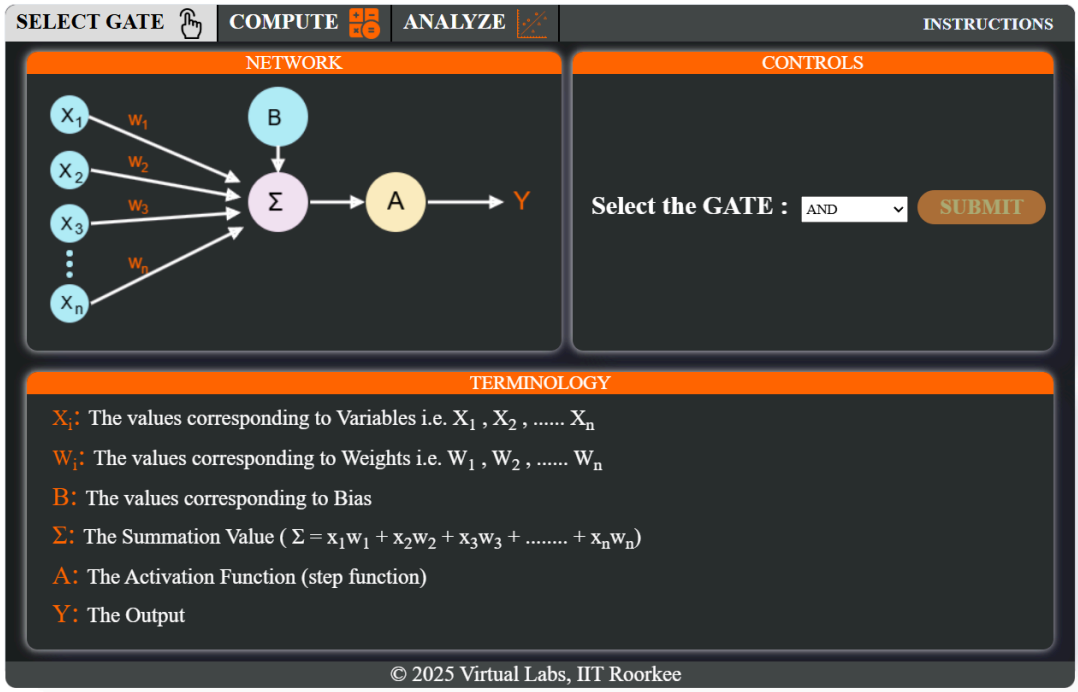
Step 4: Apply activation function

Step 5: Compare predicted output with actual output

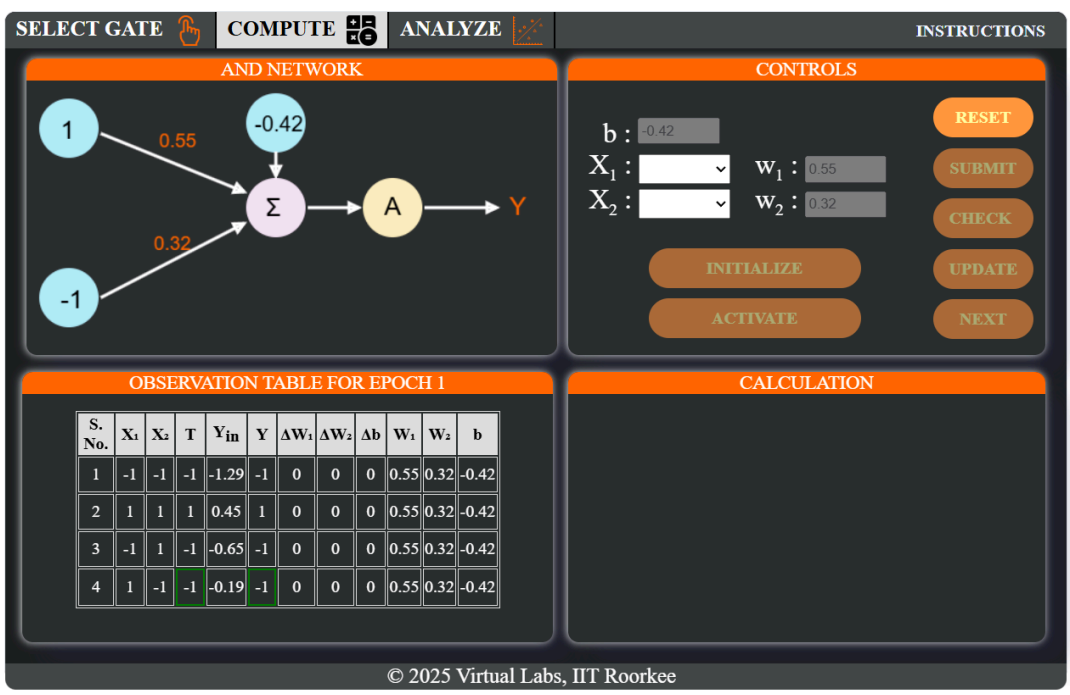
Step 6: Update weights using learning rule

Step 7: Repeat until the error becomes zero

# Initial Setup

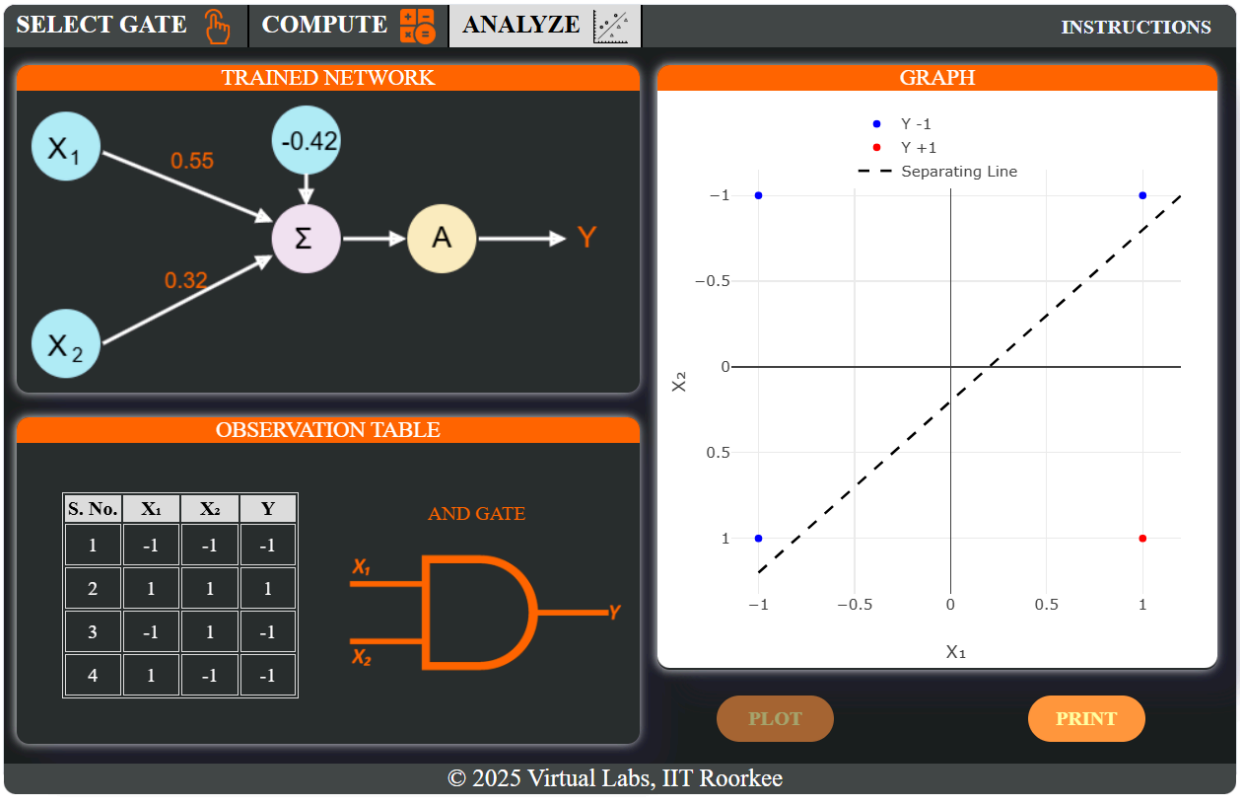



# After Computation




| S. No. | $X_1$ | $X_2$ | T | $Y_{in}$ | Y | $\Delta W_1$ | $\Delta W_2$ | $\Delta b$ | $W_1$ | $W_2$ | b |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | -1 | -1 | -1 | -1.29 | -1 | 0 | 0 | 0 | 0.55 | 0.32 | -0.42 |
| 2 | 1 | 1 | 1 | 0.45 | 1 | 0 | 0 | 0 | 0.55 | 0.32 | -0.42 |
| 3 | -1 | 1 | -1 | -0.65 | -1 | 0 | 0 | 0 | 0.55 | 0.32 | -0.42 |
| 4 | 1 | -1 | -1 | -0.19 | -1 | 0 | 0 | 0 | 0.55 | 0.32 | -0.42 |

## Graph




## Result

The Single Layer Perceptron successfully classified the given input data into binary classes.

## Conclusion

The Single Layer Perceptron is a simple and effective model for binary classification. However, it can only classify linearly separable data and fails for non-linear problems.The Single Layer Perceptron successfully classified the given input data into binary classes.